

using integration to compute $R_{2,2}$ for
following integrals:

$$\int_0^1 x^b e^{-x} dx$$

Let $f(x) = x^b e^{-x}$.
Here, $a > 0$ and $b > 1$.

Solution:

and

$$h_k = \frac{b-a}{2^{k-1}}$$

For, $k=1$, $h_1 = \frac{1-0}{2^{1-1}} = \frac{1}{1} = 1$

For, $k=2$, $h_2 = \frac{1-0}{2^{2-1}} = \frac{1}{2} = 0.5$

For, $k=3$, $h_3 = \frac{1-0}{2^{3-1}} = \frac{1}{4} = 0.25$

$$R_{1,1} = \frac{h_1}{2} [f(a) + f(b)]$$

$$= \frac{1}{2} [f(0) + f(1)] = \frac{1}{2} [0 + 0.367879]$$

$$= 0.1839397206$$

$$R_{2,2} = \frac{h_2}{2} [f(a) + f(b) + 2f(a + h_2)]$$

$$= \frac{0.5}{2} [f(0) + f(1) + 2f(0.5)]$$

$$= 0.1677811928$$

$$R_{3,1} = \frac{h_3}{2} [f(a) + f(b) +$$

$$2f(a + h_3) + f(a + 2h_3)$$

$$+ f(a + 3h_3)]$$

$$= \frac{0.25}{2} [f(0) + f(1) + 2f(0.25)$$

$$+ f(0.5) + f(0.75)]$$

$$= 0.112488405$$

improving the composit trapezoidal rule
 using richardson extrapolation

$$R_{2,2} = R_{2,1} + \frac{R_{2,1} - R_{1,1}}{4^{2-1} - 1}$$

$$= \dots = 0.16240168$$

$$R_{3,2} = R_{3,1} + \frac{R_{3,1} - R_{2,1}}{4^{2-1} - 1}$$

$$= \dots = 0.1607224759$$

$$R_{3,3} = R_{3,2} + \frac{R_{3,2} - R_{2,2}}{4^{3-1} - 1}$$

$$= \dots = 0.1606105289$$

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